



HYDROS3.8



SAMUDRAMANTHAN 2025

HIGH-SPEED PLANING HULL INTEGRATED WITH AN AZIMUTHAL PROPULSION SYSTEM FOR SUPERIOR MANEUVERABILITY. IT'S CONTROLLED VIA A 6-CHANNEL RC SYSTEM WITH PID TUNING, ENABLING PRECISION THRUST VECTORING AND AGILE NAVIGATION



IN SEARCH OF INNOVATION

BuildNEO: A student-led research and development initiative at the Indian Maritime University, Visakhapatnam, focused on innovation and practical skill development in naval architecture. Founded by ambitious alumni, it thrives on a network of past and present students collaborating on working prototypes and research. The initiative features a dynamic leadership team, including roles in external communications, electronics, technical documentation, coding, CAD modeling, fabrication, additive manufacturing, and social media. By promoting project flexibility and creativity, BuildNEO prepares members for academic and industrial challenges, aiming to become a hub of talent and innovation in the maritime industry.

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This boat is a technologically advanced marine vessel designed for superior performance, maneuverability, and stability. Incorporating modern naval architectural principles and propulsion technologies, the vessel ensures operational efficiency and adaptability across a range of marine conditions.

OBJECTIVE

The objective is to develop a stable and efficient marine vessel capable of fulfilling various operational roles by maximizing thrust efficiency and maneuverability through innovative propulsion and hull design, ensuring safety and adaptability with a dynamic ballasting system, and validating performance through rigorous testing under diverse load and environmental conditions.

SPECIFICATIONS

- 1.Length 34.5 cm
- 2.Breadth 15 cm
- 3.Draft 4.5 cm
- 4.Max speed 3.8
- 5.ENDURANC HR = 30 - 70 mintues
- 6.CAPACITY: Motor 1400kv
- 7.Azimuthal propulsion(z drive Variant
- 8.Catamaran
- 9.Propeller diameter 4cm
- 10.Controler and reciver are same that fs i6 one

FEATURES

- Catamaran Hull Form: Provides enhanced stability, reduced drag, and higher speed with improved fuel efficiency.
- Azimuthal Propulsion: Allows 360-degree thrust direction, offering exceptional maneuverability and ease of docking or navigation in tight spaces.
- Ballasting System: Enables control of buoyancy and trim, enhancing stability and performance in various load conditions.
- Load Testing and Optimization: Extensively tested under multiple load scenarios to ensure maximum maneuverability and thrust efficiency.

